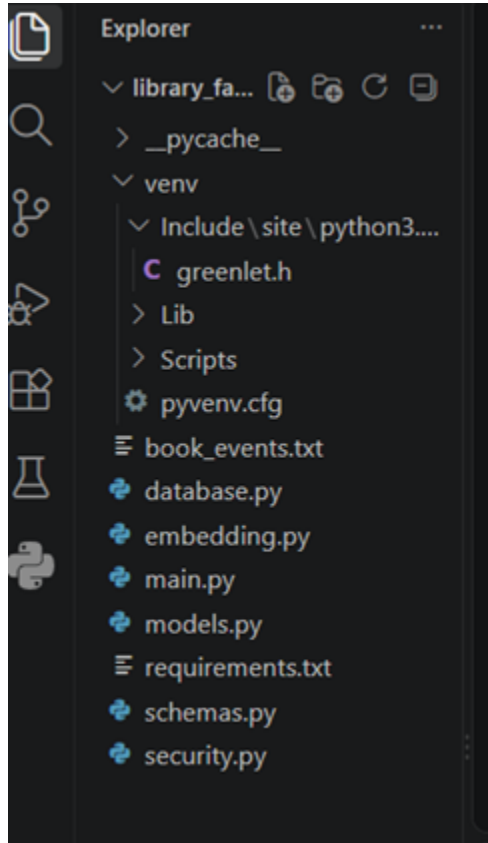
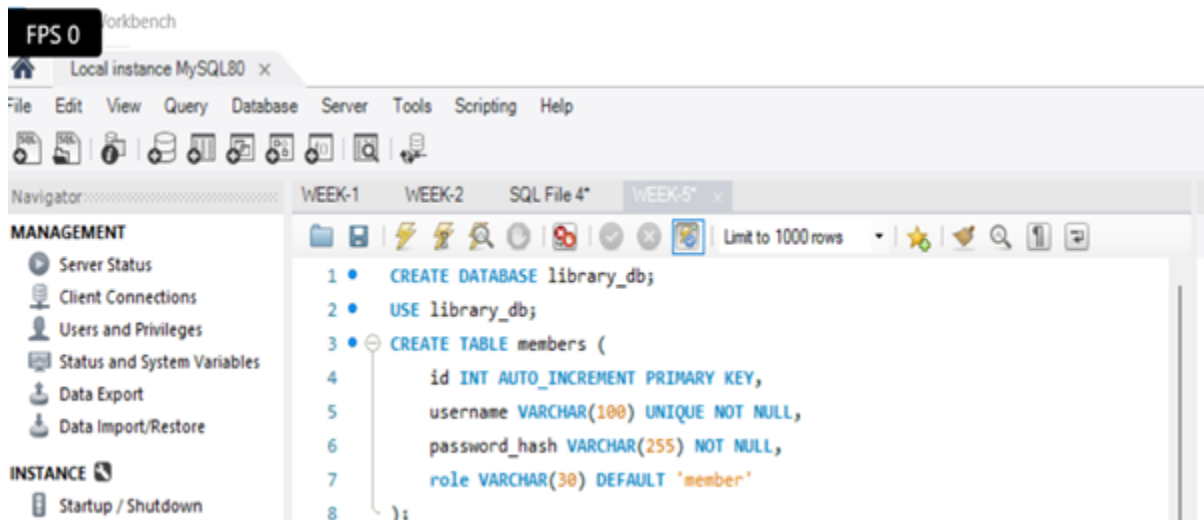


PRACTICAL WEEK-5

Project Structure



MySQL Setup



MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```
8 }
9 CREATE TABLE books (
10   id INT AUTO_INCREMENT PRIMARY KEY,
11   title VARCHAR(255) NOT NULL,
12   author VARCHAR(255) NOT NULL,
13   isbn VARCHAR(13) UNIQUE NOT NULL,
14   price DECIMAL(10,2) NOT NULL,
15   published_date DATE NOT NULL,
16   description TEXT,
17   embedding JSON
18 );
19 SHOW DATABASES;
```

Result Grid

Database

- bookflow_db
- bookflow_db
- information_schema
- kludb
- library_db
- madhulika_2520090056
- mysql
- performance_schema

Result 1 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	20:05:05	USE library_db	0 row(s) affected	0.000 sec
3	20:05:19	CREATE TABLE members (id INT AUTO_INCREMENT PRI...	0 row(s) affected	0.141 sec
4	20:05:32	CREATE TABLE books (id INT AUTO_INCREMENT PRIM...	0 row(s) affected	0.062 sec

Automatic context help is disabled. Use the tool manually get help for current caret position toggle automatic help

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```
10   id INT AUTO_INCREMENT PRIMARY KEY,
11   title VARCHAR(255) NOT NULL,
12   author VARCHAR(255) NOT NULL,
13   isbn VARCHAR(13) UNIQUE NOT NULL,
14   price DECIMAL(10,2) NOT NULL,
15   published_date DATE NOT NULL,
16   description TEXT,
17   embedding JSON
18 );
19 SHOW DATABASES;
20 USE library_db;
21 SHOW TABLES;
```

Result Grid

Tables_in_library_db

- books
- members

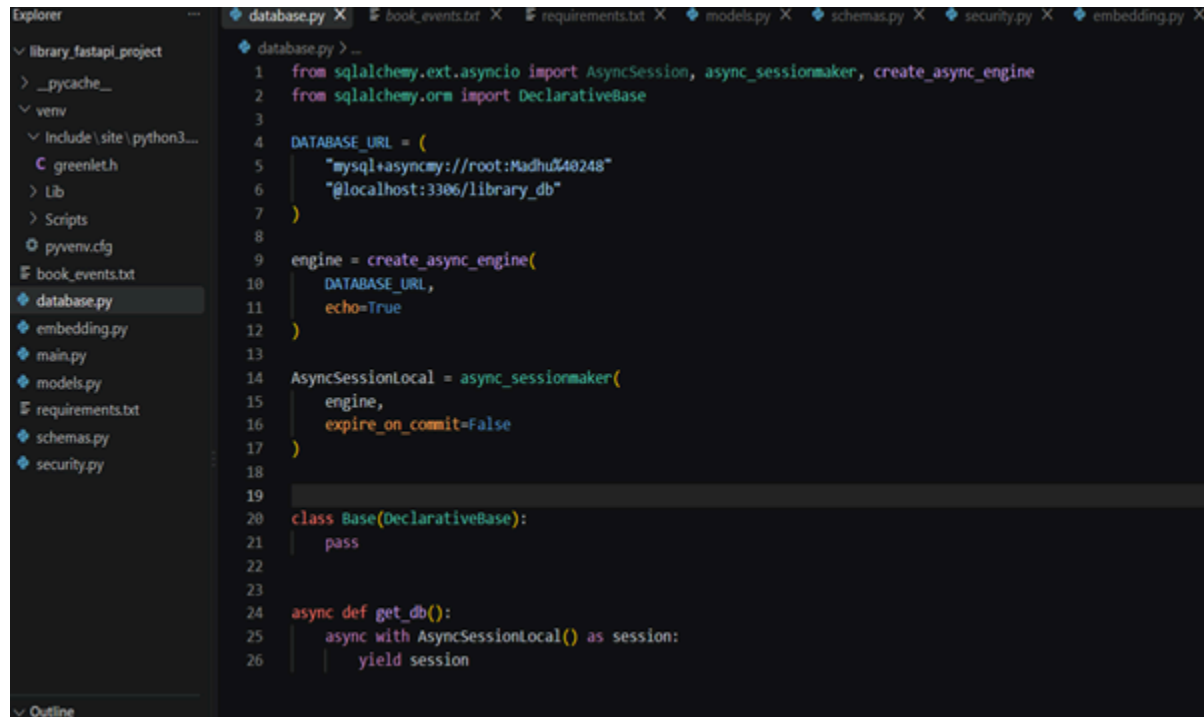
Result 2 x

Output

Action Output

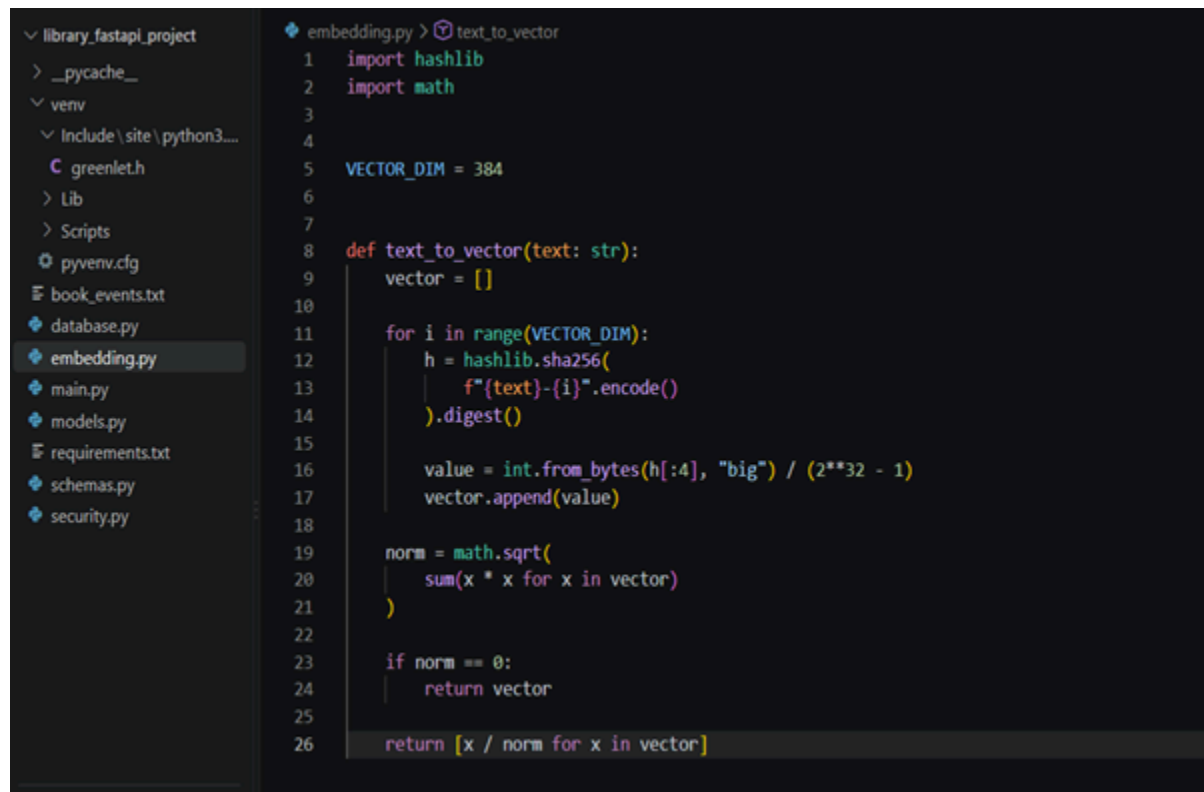
#	Time	Action	Message
4	20:05:32	CREATE TABLE books (id INT AUTO_INCREMENT PRIM...	0 row(s) affected
5	20:05:55	SHOW DATABASES	10 row(s) returned
6	20:07:46	USE library_db	0 row(s) affected
7	20:07:52	SHOW TABLES	2 row(s) returned

Database.py



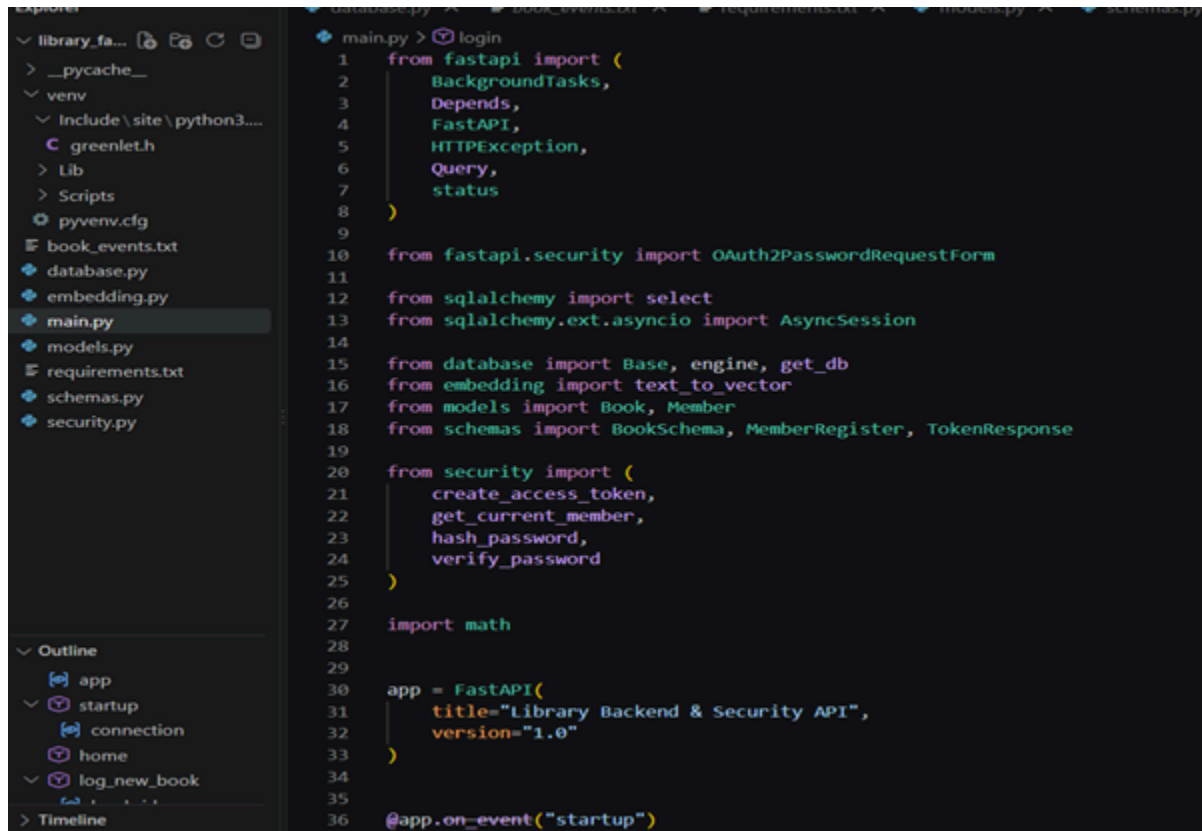
```
1 from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker, create_async_engine
2 from sqlalchemy.orm import DeclarativeBase
3
4 DATABASE_URL = (
5     "mysql+asyncmy://root:Madhu%40248"
6     "@localhost:3306/library_db"
7 )
8
9 engine = create_async_engine(
10     DATABASE_URL,
11     echo=True
12 )
13
14 AsyncSessionLocal = async_sessionmaker(
15     engine,
16     expire_on_commit=False
17 )
18
19
20 class Base(DeclarativeBase):
21     pass
22
23
24 async def get_db():
25     async with AsyncSessionLocal() as session:
26         yield session
```

Embedding.py



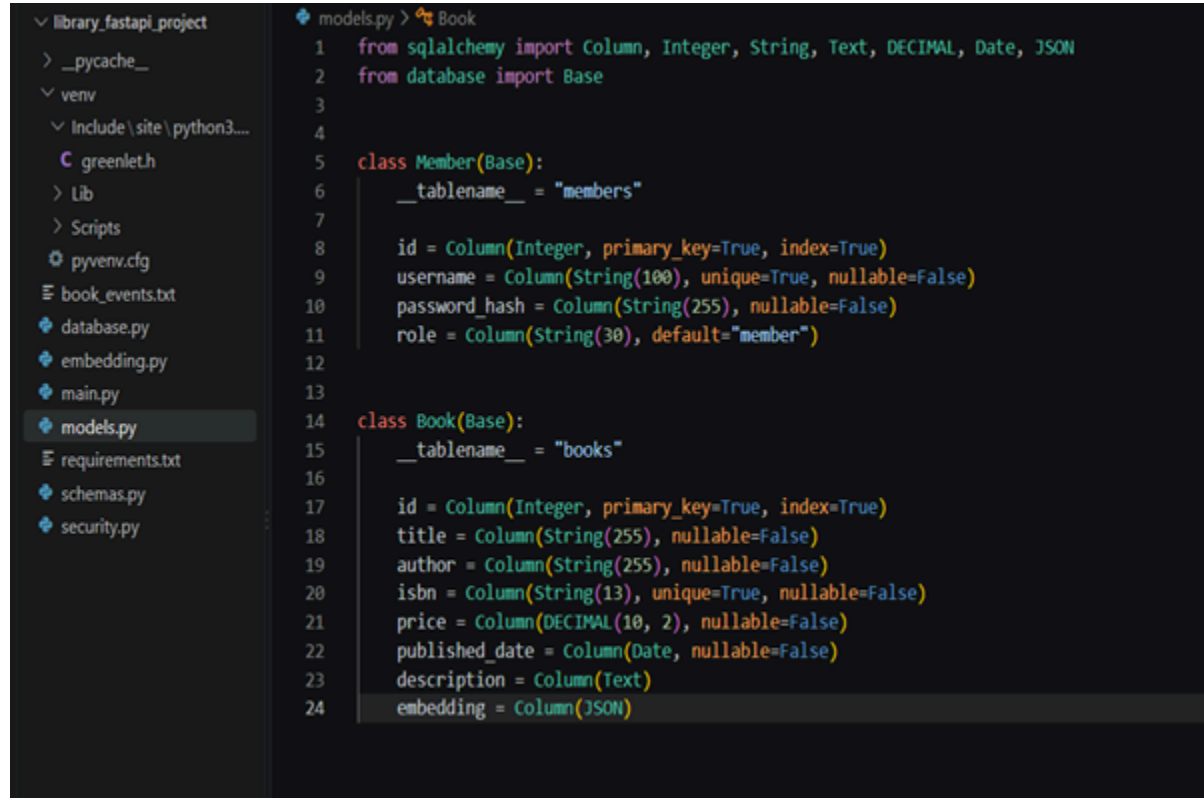
```
1 import hashlib
2 import math
3
4
5 VECTOR_DIM = 384
6
7
8 def text_to_vector(text: str):
9     vector = []
10
11     for i in range(VECTOR_DIM):
12         h = hashlib.sha256(
13             f"{text}-{i}".encode()
14         ).digest()
15
16         value = int.from_bytes(h[:4], "big") / (2**32 - 1)
17         vector.append(value)
18
19     norm = math.sqrt(
20         sum(x * x for x in vector)
21     )
22
23     if norm == 0:
24         return vector
25
26     return [x / norm for x in vector]
```

Main.py



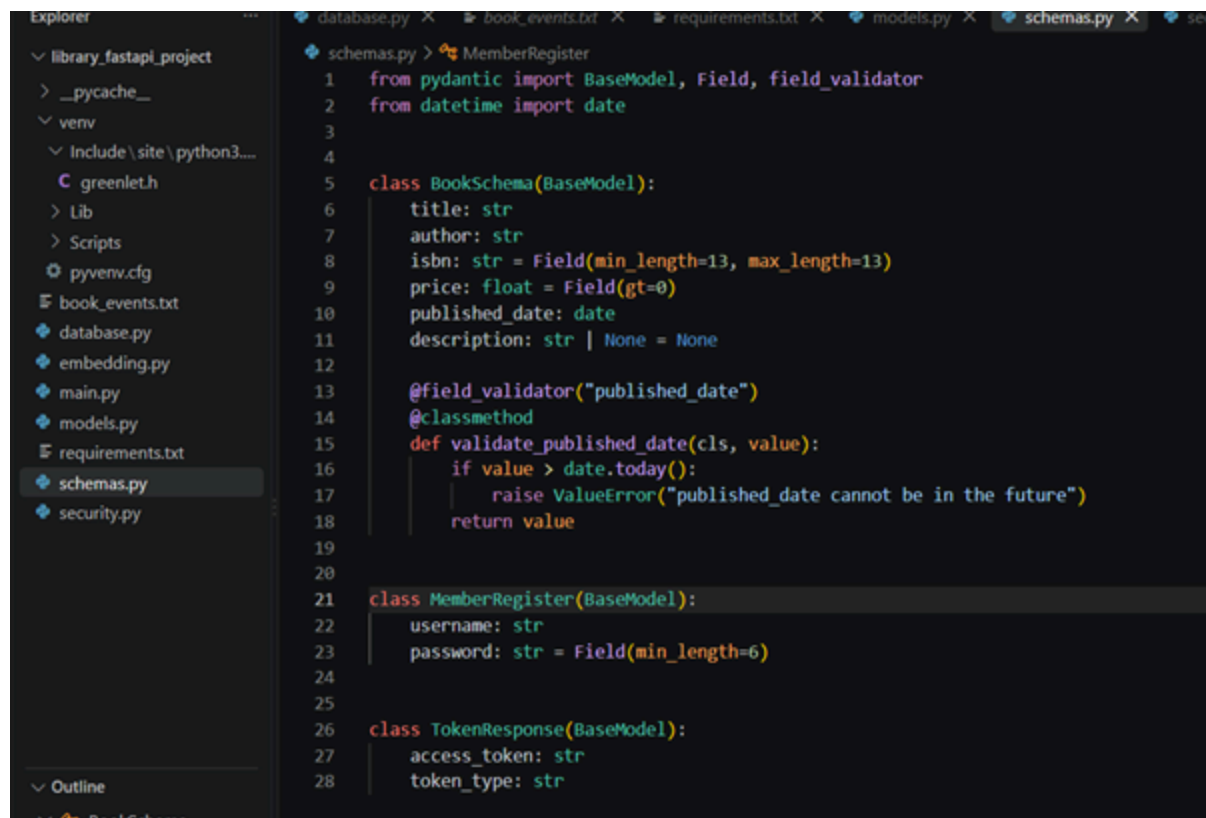
```
1 from fastapi import (
2     BackgroundTasks,
3     Depends,
4     FastAPI,
5     HTTPException,
6     Query,
7     status
8 )
9
10 from fastapi.security import OAuth2PasswordRequestForm
11
12 from sqlalchemy import select
13 from sqlalchemy.ext.asyncio import AsyncSession
14
15 from database import Base, engine, get_db
16 from embedding import text_to_vector
17 from models import Book, Member
18 from schemas import BookSchema, MemberRegister, TokenResponse
19
20 from security import (
21     create_access_token,
22     get_current_member,
23     hash_password,
24     verify_password
25 )
26
27 import math
28
29
30 app = FastAPI(
31     title="Library Backend & Security API",
32     version="1.0"
33 )
34
35
36 @app.on_event("startup")
```

Models.py



```
1 from sqlalchemy import Column, Integer, String, Text, DECIMAL, Date, JSON
2 from database import Base
3
4
5 class Member(Base):
6     __tablename__ = "members"
7
8     id = Column(Integer, primary_key=True, index=True)
9     username = Column(String(100), unique=True, nullable=False)
10    password_hash = Column(String(255), nullable=False)
11    role = Column(String(30), default="member")
12
13
14 class Book(Base):
15     __tablename__ = "books"
16
17     id = Column(Integer, primary_key=True, index=True)
18     title = Column(String(255), nullable=False)
19     author = Column(String(255), nullable=False)
20     isbn = Column(String(13), unique=True, nullable=False)
21     price = Column(DECIMAL(10, 2), nullable=False)
22     published_date = Column(Date, nullable=False)
23     description = Column(Text)
24     embedding = Column(JSON)
```

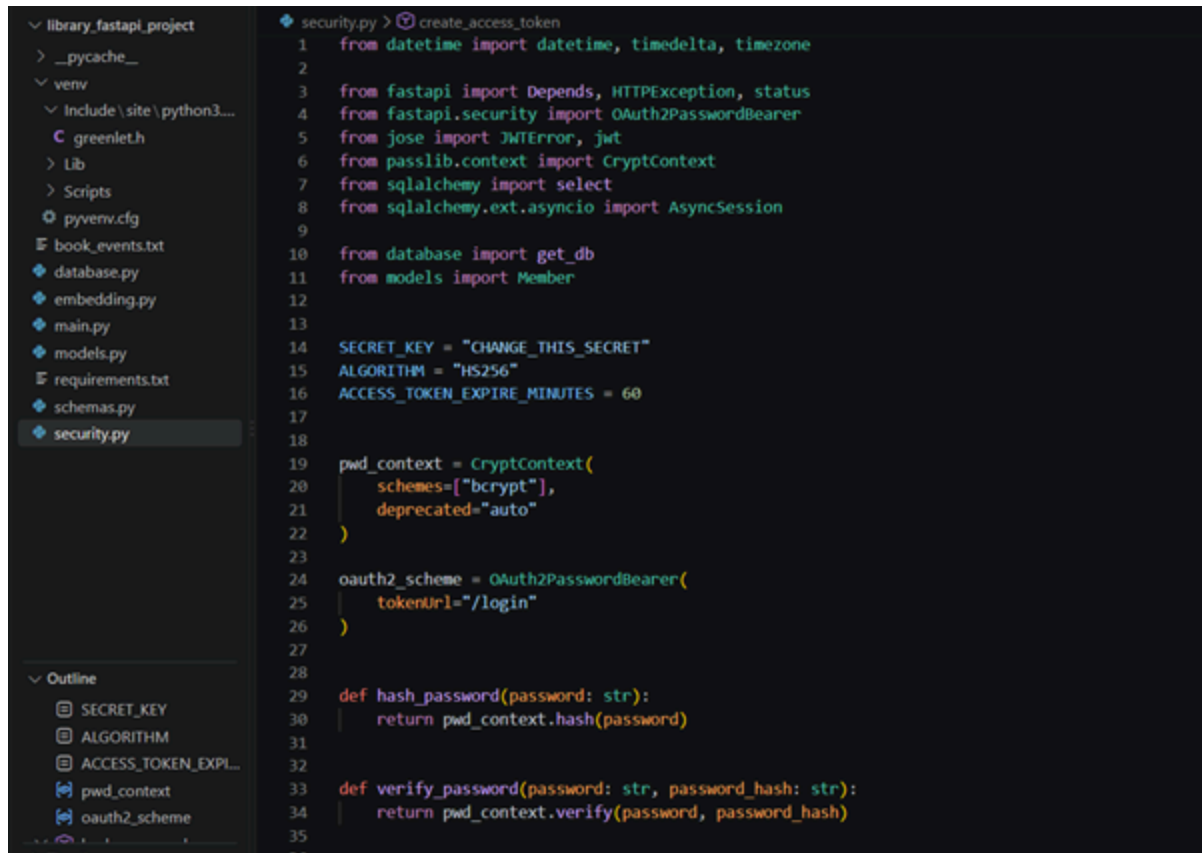
Schemas.py



The image shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'library_fastapi_project' with several files and folders. The code editor shows the 'schemas.py' file with the following code:

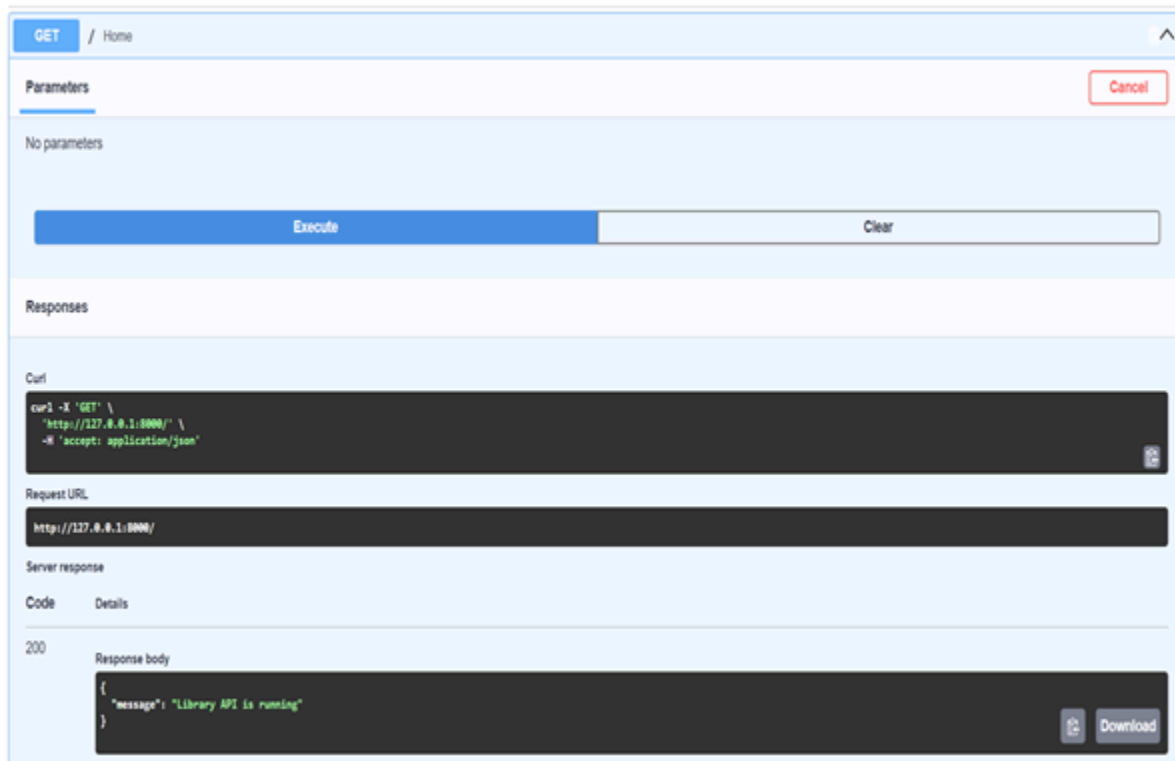
```
1 from pydantic import BaseModel, Field, field_validator
2 from datetime import date
3
4
5 class BookSchema(BaseModel):
6     title: str
7     author: str
8     isbn: str = Field(min_length=13, max_length=13)
9     price: float = Field(gt=0)
10    published_date: date
11    description: str | None = None
12
13    @field_validator("published_date")
14    @classmethod
15    def validate_published_date(cls, value):
16        if value > date.today():
17            raise ValueError("published_date cannot be in the future")
18        return value
19
20
21 class MemberRegister(BaseModel):
22     username: str
23     password: str = Field(min_length=6)
24
25
26 class TokenResponse(BaseModel):
27     access_token: str
28     token_type: str
```

Security.py



```
security.py > create_access_token
1  from datetime import datetime, timedelta, timezone
2
3  from fastapi import Depends, HTTPException, status
4  from fastapi.security import OAuth2PasswordBearer
5  from jose import JWTError, jwt
6  from passlib.context import CryptContext
7  from sqlalchemy import select
8  from sqlalchemy.ext.asyncio import AsyncSession
9
10 from database import get_db
11 from models import Member
12
13
14 SECRET_KEY = "CHANGE_THIS_SECRET"
15 ALGORITHM = "HS256"
16 ACCESS_TOKEN_EXPIRE_MINUTES = 60
17
18
19 pwd_context = CryptContext(
20     schemes=["bcrypt"],
21     deprecated="auto"
22 )
23
24 oauth2_scheme = OAuth2PasswordBearer(
25     tokenUrl="/login"
26 )
27
28
29 def hash_password(password: str):
30     return pwd_context.hash(password)
31
32
33 def verify_password(password: str, password_hash: str):
34     return pwd_context.verify(password, password_hash)
35
36
```

Test Registration



GET / Home

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/

Server response

Code	Details
200	Response body <pre>{ "message": "Library API is running" }</pre> Download

POST

/register

Register

^

Parameters

Cancel

Reset

No parameters

Request body

required

application/json

▼

Edit Value

Schema

```
{
  "username": "Radhu",
  "password": "Library123"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Radu",
    "password": "Library123"
  }'
```

Request URL

```
http://127.0.0.1:8000/register
```

Server response

Code	Details
200	Response body <pre>{ "message": "Member registered successfully", "username": "Radu", "role": "member" }</pre> Response headers <pre>content-length: 79 content-type: application/json date: Mon, 14 Sep 2025 16:05:16 GMT server: uvicorn</pre>

Responses

Test Login

POST

/login Login

^

Parameters

Cancel

Reset

No parameters

Request body ^{required}

application/x-www-form-urlencoded ▾

grant_type

string | (string | null)

pattern: "password"

password

☐ Send empty value

username ^{required}

string

Madhu

☐ Send empty value

password ^{required}

string(\$password)

☐ Send empty value

scope

string

scope

☒ Send empty value

client_id

string | (string | null)

Madhu

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value


```
{
  "url": "http://127.0.0.1:8000/books/",
  "method": "POST",
  "headers": {
    "accept": "application/json",
    "authorization": "Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIjOiJlbnVpdSIsImF1dGUiOiJ1bWVudDQyYm9keSIsImV4cCI6MTkxOTMwMDAwfQ.YT9.Lq?e=7880XyQ8:IEqTt64r7GZ1shu43g8nE1H",
    "content-type": "application/json"
  },
  "body": {
    "title": "Python for Beginners",
    "author": "John Smith",
    "isbn": "123",
    "price": 450.00,
    "published_date": "2024-06-18",
    "description": "An introduction to Python programming."
  }
}
```

Request URL

<http://127.0.0.1:8080/books>

Server response

Code	Details
------	---------

422 Error: Unprocessable Entity

Response body

```
{
  "detail": {
    {
      "type": "string_too_short",
      "loc": [
        "body",
        "isbn"
      ],
      "msg": "String should have at least 13 characters",
      "input": "123",
      "ctx": {
        "min_length": 13
      }
    }
  ]
}
```

Response headers

```
content-length: 110
content-type: application/json
date: Mon, 14 Sep 2026 16:20:41 GMT
server: uvicorn
```

GET

/books/search Search Books

Cancel

Parameters

Name	Description
q required	
string (query)	q = python minLength: 1
limit	
integer (query)	5 maxLength: 20 minLength: 1

ExecuteClear

Responses

Curl

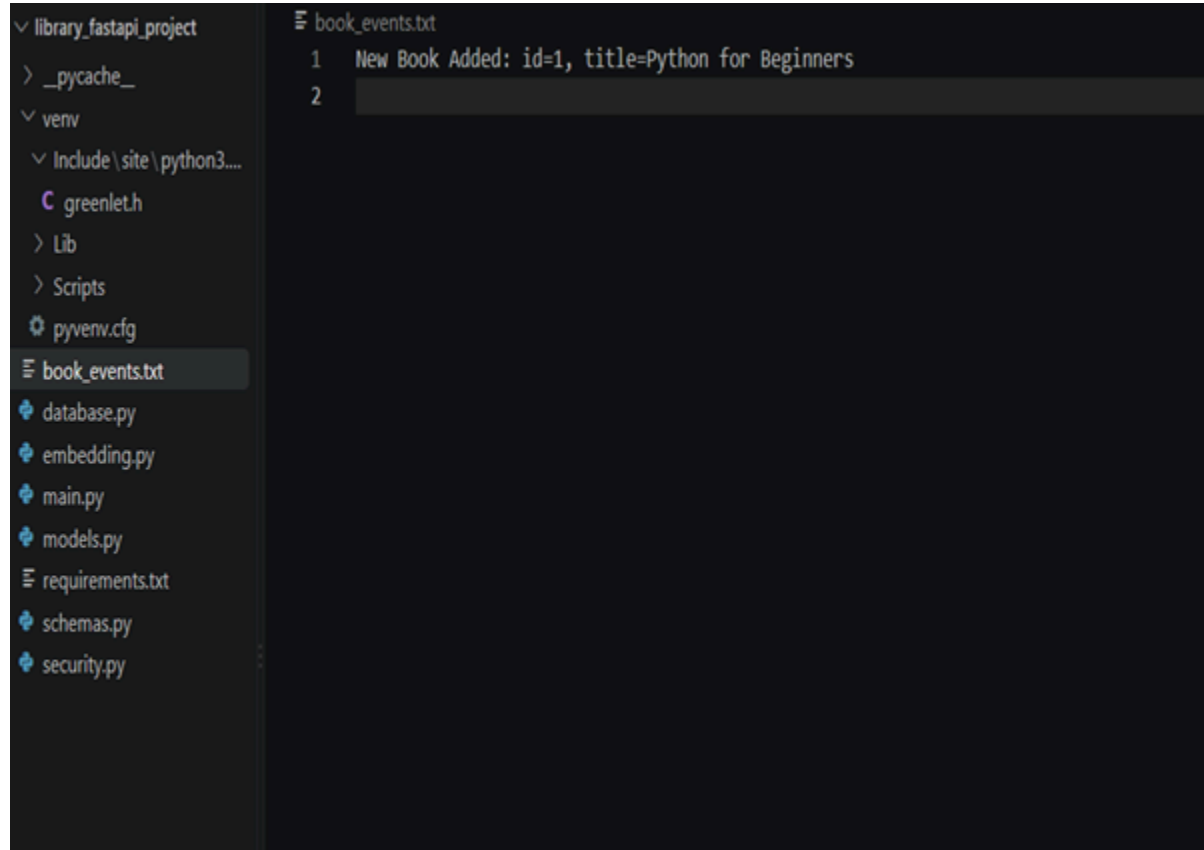
```
curl -X 'GET' \
  'http://127.0.0.1:8080/books/search?q=q&limit=5' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/books/search?q=q&limit=5
```

Server response

book_event.txt



```
library_fastapi_project
├── __pycache__
├── venv
│   ├── Include\site\python3....
│   ├── greenlet.h
│   ├── Lib
│   ├── Scripts
│   ├── pyvenv.cfg
│   └── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
└── security.py
```

```
book_events.txt
1 New Book Added: id=1, title=Python for Beginners
2
```